

Experimental Analysis of Public Policies

Sessions 2–3

Selection, Matching, Randomised Experiments, and Power

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Session Objectives

By the end of this session, students should be able to:

- separate description, prediction, and causal interpretation in observational comparisons;
- explain how regression and matching use observed covariates to construct a counterfactual;
- state the assumptions and diagnostics behind matching, propensity scores, overlap, and balance;
- explain how randomisation changes identification, design choices, implementation risks, and power.

Roadmap

Causal Objects and Selection

Regression and Controls

Matching and Propensity Scores

Randomised Identification

Designing the Experiment

Implementation Threats

Synthesis

From Policy Question to Counterfactual

Policy question

- Does a job-search workshop improve employment?
- For whom: all eligible workers or current participants?
- At what horizon: six months, one year, longer?

Causal comparison

$$Y_i(1) - Y_i(0)$$

- Same worker.
- Same outcome and horizon.
- Two treatment states, only one observed.

Course reflex

Where does the identifying variation come from? What assumption makes the comparison causal? Which population does the estimate describe?

Rubin Causal Model: The Evaluation Objects

Objects

- Unit: worker i .
- Treatment: workshop participation $D_i \in \{0, 1\}$.
- Potential outcomes: $Y_i(1)$, $Y_i(0)$.
- Observed outcome:
$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$$

Design conditions

- Consistency: the treatment version must be defined.
- $Y_i(1)$ and $Y_i(0)$ refer to the same outcome and horizon.
- No interference: one worker's treatment should not change another worker's potential outcomes.
- The assignment mechanism determines which potential outcome is observed.

Fundamental problem

For the same worker at the same time, one potential outcome is observed and the other is counterfactual.

Target Populations Determine the Estimand

Policy margin

- Evaluate the workshop for current participants.
- Expand access to all eligible workers.
- Retarget current non-participants.

Average effects

Estimand		Population
ATE	=	all eligible workers
$E[Y_i(1)$	–	
$Y_i(0)]$		
ATT	=	workshop participants
$E[Y_i(1)$	–	
$Y_i(0)$		
$D_i = 1]$		
ATU	=	non-participants
$E[Y_i(1)$	–	
$Y_i(0)$		
$D_i = 0]$		

Description, Prediction, and Causality Answer Different Questions

Task	Question	What it can support
Description	Are workshop participants more often employed?	an observed association
Prediction	Which variables forecast employment?	expected outcomes for similar workers
Causality	What would the same workers have experienced without the workshop?	a counterfactual effect under an identifying assumption

A model can predict well and still fail as a causal design if it does not justify the missing counterfactual.

Running Example: Job-Search Workshop

Observed data

- Treatment $D_i = 1$: attends the workshop.
- Outcome Y_i : employed after six months.
- Baseline X_i : age, education, unemployment duration, prior earnings, sector, local office.

Selection channels

- Motivation.
- Informal networks.
- Health constraints.
- Caseworker referral rules.

The Observed Gap Mixes Effect and Selection

	<i>N</i>	Employed	Mean
Participants	100	55	0.55
Non-participants	100	45	0.45
Gap			0.10

$$\Delta^{naive} = E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]$$

$$\begin{aligned}\Delta^{naive} &= ATT \\ &\quad + \{E[Y_i(0) \mid D_i = 1] \\ &\quad - E[Y_i(0) \mid D_i = 0]\}\end{aligned}$$

Interpretation

The 10 percentage-point gap is causal only if controls represent participants' missing untreated outcomes.

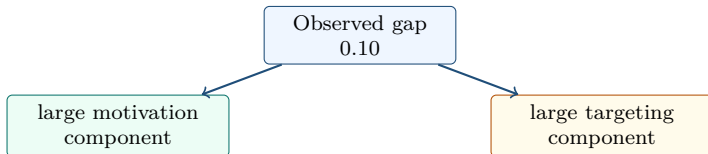
The Same Gap Can Reflect Opposite Selection Stories

Positive selection

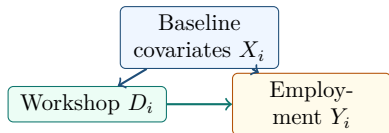
Motivated workers choose the workshop. Their untreated outcomes would already be higher. The naive gap is biased upward.

Negative selection

Caseworkers target harder-to-place workers. Their untreated outcomes would already be lower. The naive gap is biased downward.



Confounders Create Non-Comparable Groups



- A confounder affects treatment and potential outcomes.
- Controls help only if they capture the relevant selection channels.
- Unobserved confounders remain outside the adjustment.

Conditioning on Observables Reconstructs a Counterfactual

$$E[Y_i(0) \mid D_i = 1, X_i] = E[Y_i(0) \mid D_i = 0, X_i].$$

What conditioning tries to do

- Compare workers with similar X_i .
- Remove observed selection channels.
- Use untreated workers as local counterfactuals.

What it must assume

- Relevant confounders are measured.
- Covariates are pre-treatment.
- Treated and untreated workers overlap.

Regression Moves from a Naive to a Conditional Comparison

No controls

$$Y_i = \alpha + \beta D_i + \varepsilon_i$$

β is the mean outcome gap between participants and non-participants.

With controls

$$Y_i = \alpha + \beta D_i + X_i' \gamma + \varepsilon_i$$

β is an adjusted gap among workers with similar included covariates.

Interpretation

What changes is the comparison group. What does not change is the need for an identifying assumption.

The Coefficient Is Causal Only Under Conditional Comparability

Regression comparison	Needed for causality	Main failure mode
No controls	treated and untreated groups are comparable without adjustment	selection into participation
Controls X_i	all important pre-treatment confounders are included and measured well	unobserved motivation, networks, health
Controls plus baseline outcome	baseline outcome captures much prior risk	post-treatment variables or poor overlap

Omitted-Variable Bias Has Two Ingredients

Suppose the causal model includes a pre-treatment factor W_i :

$$Y_i = \alpha + \beta D_i + \delta W_i + u_i.$$

If W_i is omitted from a simple regression:

$$\tilde{\beta} = \beta + \delta \frac{\text{Cov}(D_i, W_i)}{\text{Var}(D_i)}.$$

Bias requires

- W_i affects Y_i .
- W_i is correlated with D_i .

Sign rule

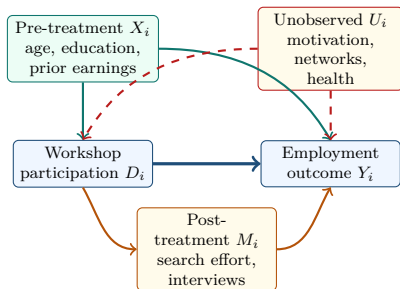
- Same sign: upward bias.
- Opposite signs: downward bias.

Bias Direction in the Workshop Example

Omitted factor	Relation to treatment and outcome	Bias
Motivation	raises participation and employment	upward
Long unemployment duration	raises referral, lowers employment	downward
Local labour-market strength	affects employment and referral rules	depends on assignment story

Coefficient $\rightarrow$ selection story: what is omitted, how it affects Y_i , and how it relates to D_i .

Which Variables Should We Control For?



Use?	Reason
Control	pre-treatment variables that affect D_i and Y_i
Avoid	variables caused by the workshop, because they can absorb the effect
Cannot fix	unobserved selection channels remain a threat

Controls Can Improve Precision Without Solving Identification

Identification

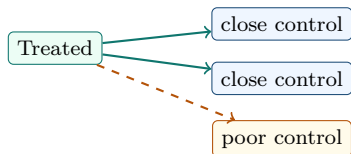
- Is the comparison causal?
- Are potential outcomes comparable after conditioning?
- This is a design claim.

Precision

- How uncertain is the estimate?
- Do covariates explain residual outcome variation?
- This is an inference gain.

$$Y_i = \alpha + \tau D_i + X_i' \gamma + \varepsilon_i$$

Matching Makes the Comparison Group Visible



- Start from treated workers.
- Find untreated workers with similar pre-treatment X_i .
- Estimate the outcome gap in the matched comparison.
- Inspect bad matches and missing support.

Exact Matching Is Transparent but Fragile

Worker	D_i	Age	Educ.	Unemp.	Y_i
Amina	1	24	low	long	1
David	0	25	low	long	0
Boris	1	45	high	short	1
Elise	0	44	high	short	1
Chloe	1	32	low	short	0
Grace	0	36	low	short	0

- Exact matches are easy to defend.
- Continuous variables make exact matches scarce.
- Adding many covariates can destroy support.

The Matched Comparison Changes the Counterfactual

Naive comparison

100 treated vs 100 controls

$$\hat{\Delta}^{naive} = 0.55 - 0.45 = 0.10$$

Matched comparison

Amina-David

Boris-Elise

Chloe-Grace

$$\hat{\Delta}^{matched} = \bar{Y}_T - \bar{Y}_{C(M)}$$

where $C(M)$ denotes matched controls.

$$\text{Toy matched gap} = 2/3 - 1/3 = 1/3.$$

Remaining limitation

Matching changes the observed comparison group.

It does not remove unobserved motivation, networks, or health constraints.

Nearest Neighbours and Replacement Are Match-Quality Choices

Choice	Purpose	Risk
Nearest neighbour	choose untreated units closest in X_i or a score	nearest can still be far away
Caliper	reject matches beyond a maximum distance	fewer treated units remain
With replacement	allow best controls to be reused	high weight on a few controls
Without replacement	use each control once	later treated units may get worse matches

ATT Matching Estimator

Let w_{ij} be the weight placed on untreated worker j as a match for treated worker i :

$$\widehat{ATT}_M = \frac{1}{N_T} \sum_{i:D_i=1} \left(Y_i - \sum_{j:D_j=0} w_{ij} Y_j \right).$$

- The target is often the ATT.
- The comparison group is built unit by unit.
- Bad support narrows the treated population.
- Standard errors must reflect matching or weighting.

Propensity Scores Reduce the Dimension of Matching

$$e(X_i) = \Pr(D_i = 1 \mid X_i)$$

Problem

- Many covariates.
- Continuous variables.
- Sparse exact cells.

Score

- Summarises observed selection into treatment.
- Groups workers with similar predicted treatment probability.
- Must be judged by balance, not model fit alone.

Propensity Scores Reduce Dimension, Not Assumptions

Conditional independence

$$(Y_i(0), Y_i(1)) \perp D_i \mid X_i$$

Observed covariates are sufficient for comparability.

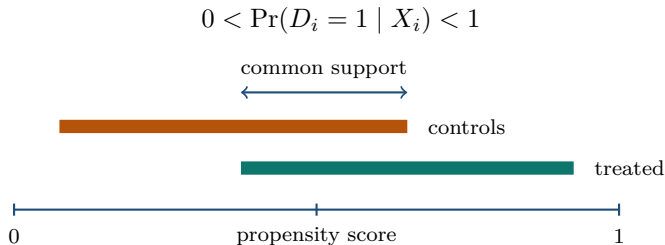
Balancing score

If $e(X_i)$ is the true propensity score:

$$D_i \perp X_i \mid e(X_i)$$

If conditional independence holds given X_i , it also holds given $e(X_i)$.

Common Support Defines Where Matching Is Credible



Outside common support, the estimate relies on extrapolation or the target population must be narrowed.

Balance Is Checked Before Interpreting Outcomes

Before matching

- Treated workers have longer unemployment spells.
- Control workers include many low-risk cases.
- The outcome gap mixes treatment and baseline differences.

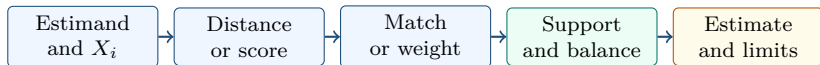
After matching

- Covariate distributions should look similar.
- Standardised differences should shrink.
- Balance on X_i is not balance on unobservables.

Order of operations

Design the match, check support and balance, then interpret the outcome contrast.

Matching Workflow



Regression and Matching Solve the Same Problem Differently

	Regression		Matching
Comparison	adjusted mean	conditional	explicitly similar workers
Main choice	controls	and functional form	distance, score, weights, caliper
Diagnostic	specification	and sensitivity	balance and common support
Shared assumption	observed covariates are sufficient for comparability		
Useful combination	match or weight first, then adjust remaining differences with regression		

Observational Adjustment Has a Clear Boundary

$$(Y_i(0), Y_i(1)) \perp D_i \mid X_i$$

What diagnostics can show

- Similar observed covariates.
- Common support.
- Sensitivity to adjustment choices.

What diagnostics cannot show

- Absence of unobserved motivation.
- No hidden networks.
- No unmeasured health constraints.

Midpoint: From Argued Comparability to Created Comparability

Observational adjustment

Comparability is argued using observed X_i .

Regression and matching ask whether covariates reconstruct the missing counterfactual.

Randomisation

Comparability is created by random Z_i .

The assignment mechanism creates treated and control groups before outcomes are realised.

observed X_i $\xrightarrow{\text{turning point}}$ random Z_i

Randomisation Breaks the Link Between Assignment and Potential Outcomes

$$Z_i \perp (Y_i(0), Y_i(1))$$

What randomisation solves

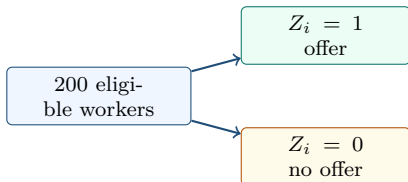
- Observed selection.
- Unobserved selection.
- Baseline comparability in expectation.

What it does not solve

- Spillovers.
- Attrition.
- Non-compliance.
- Weak power.

Workshop Lottery Creates the Comparison Group

- 200 eligible workers.
- 100 available workshop places.
- The office randomly offers access to half.
- Scarcity makes the lottery administratively meaningful.



A Difference in Means Is Causal Under Random Assignment

Assume full compliance, so $D_i = Z_i$:

$$E[Y_i(0) \mid Z_i = 1] = E[Y_i(0) \mid Z_i = 0]$$

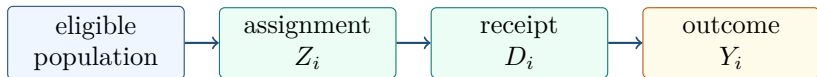
$$E[Y_i(1) \mid Z_i = 1] = E[Y_i(1) \mid Z_i = 0].$$

Therefore:

$$E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0] = E[Y_i(1) - Y_i(0)].$$

The equation is justified by the assignment mechanism, not by a richer regression specification.

Assignment, Receipt, and Outcomes



$$SATE = \frac{1}{N} \sum_{i=1}^N \{Y_i(1) - Y_i(0)\}$$

Full compliance: $D_i = Z_i$.

$$ITT = E[Y_i | Z_i = 1] \\ - E[Y_i | Z_i = 0]$$

Imperfect compliance: assignment effect first.

RCT Regression Estimates Assigned-Arm Contrasts

$$Y_i = \alpha + \tau Z_i + \varepsilon_i$$

$$Y_i = \alpha + \tau Z_i + X_i' \gamma + \lambda_s + \varepsilon_i$$

- Z_i is randomised assignment.
- τ is an assigned-arm contrast.
- X_i can improve precision.
- λ_s reflects stratified assignment.
- With imperfect compliance, τ is an ITT.

Experimental Protocol Checklist

1. Eligibility rule.
2. Unit of randomisation.
3. Treatment and control conditions.
4. Assignment probabilities.
5. Primary outcome and date.
6. Strata or clusters.
7. Main estimating equation.
8. Attrition and compliance plan.

Implementation standard

Another team should be able to reproduce the assignment rule from the protocol.

Three Randomisation Designs

Design		Mechanism	When useful
Simple		randomise each eligible unit with probability p	large samples, individual delivery, low spillovers
Stratified blocked	or	randomise within pre-defined groups	small samples, strong baseline predictors, local-office balance
Clustered		randomise offices, schools, villages, or regions	group delivery, contamination risk, administrative constraints

Choose the Randomisation Unit to Match Delivery and Spillovers

Unit	Useful when	Main concern
Individual	treatment is private and spillovers are small	contamination
Caseworker	staff practices are treated	few clusters
Office	programme is delivered office-wide	power and external validity
Region	policy is territorial	few units and many shocks

More individual observations do not compensate for too few independent assignment units in a clustered design.

Balance Tables Diagnose the Realised Assignment

Report

- sample size by arm;
- control mean;
- treatment-control difference;
- standardised difference.

Do not overread

- Balance tests do not prove randomisation.
- Small samples can have important imbalances.
- Large samples can flag tiny differences.

Diagnostic rather than proof

Validity comes from the assignment process; balance tables document the realised sample.

Outcomes and Pre-Specification Link the Estimate to the Decision

Outcome choice

- Primary outcome: employment after six months.
- Mechanism: number of applications.
- Secondary outcome: earnings after one year.

Pre-specification

- sample;
- primary outcome;
- main contrast;
- covariates and strata;
- treatment of missing outcomes.

Ethics and Feasibility Shape the Estimand

Ethical feasibility

- Scarce places can justify a lottery.
- Delayed treatment may be acceptable.
- Exclusion rules must be transparent.

Econometric implication

- Feasible assignment determines Z_i .
- Z_i determines the primary contrast.
- The contrast determines the estimand.

Randomisation Solves Selection but Not Implementation

Before implementation

$$Z_i \perp (Y_i(0), Y_i(1))$$

Treatment and control groups are comparable in expectation.

After implementation starts

The realised comparison can be threatened by spillovers, attrition, non-compliance, and measurement changes.

Implementation Threats and Responses

Threat	Problem	Design response
Spillovers	controls affected by treated workers	cluster, measure exposure, define spillover estimands
Attrition	outcome observed for a selected sample	tracking, attrition table, bounds, sensitivity
Non-compliance	$D_i \neq Z_i$	report ITT and first stage; use IV logic later
Measurement changes	outcome measured differently by arm	common instruments, timing, audit trail

Bias, Precision, and Power Are Different Problems

Issue	Question	Main tool
Bias	Is the comparison causal?	design and assumptions
Precision	How uncertain is the estimate?	standard errors
Power	Would a meaningful effect likely be detected?	ex ante sample design
External validity	Where else does the result apply?	population and setting analysis

Design Levers for Precision

Reduce standard errors

- more independent units;
- balanced assignment shares;
- baseline outcomes;
- strong pre-treatment predictors.

Improve the design

- stratify on outcome predictors;
- reduce measurement error;
- increase clusters when assignment is clustered;
- align analysis with randomisation.

Method Map

Approach	Variation	Main assumption	Diagnostic
Naive gap	observed treatment status	treated and untreated are comparable	baseline differences
Regression	residual variation after controls	conditional independence and good controls	sensitivity
Matching	matched weighted comparisons or com-	conditional independence and overlap	balance, support
RCT	random assignment Z_i	assignment implemented as designed	protocol, balance

Each method is a way to justify a counterfactual comparison; none should be read as a command that automatically produces causality.

Estimand and Population Can Change Across Methods

Observational adjustment

- Regression: population depends on model support.
- Matching: trimming can focus on common support.
- ATT is often more credible than ATE when treated units define the match.

Randomised experiment

- SATE for the experimental sample.
- ITT when assignment differs from receipt.
- External validity requires a separate population argument.